

NB33: KANSER Advanced Feature Engineering

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Bu rapor, KANSER paneli için literatur destekli gelişmiş feature engineering çalışmalarını içerir. NB32 champion reçetesi (P9_REVERSE_6040 + CatBoost + 4-FE, Boot-F1=0.7300) baseline olarak alınmış, 6 feature grubu (toplam 21 feature) sistematik ablasyonla test edilmistir.

Feature Grupları:

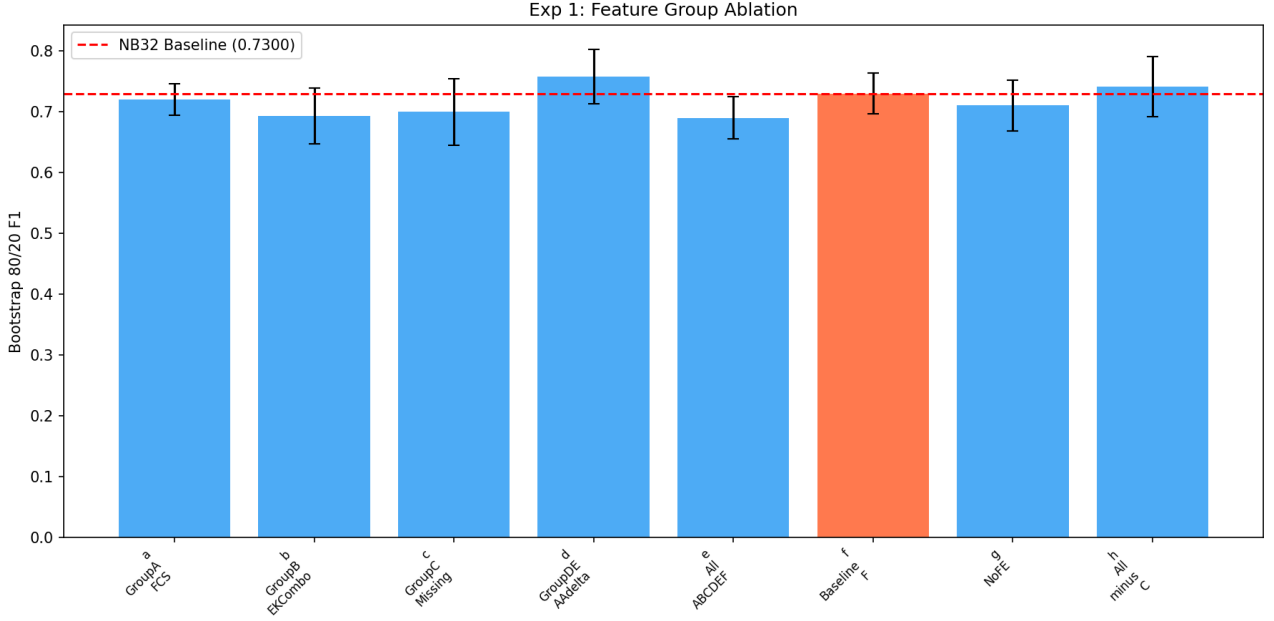
- A: FCS-tarzi Frekans x Konservasyon (3 feature)
- B: EK Skor Birlesimi / Meta-Prediktor (5 feature)
- C: Missingness / Gözlenme Sayısı (3 feature)
- D: AA Fizikokimyasal Delta (4 feature)
- E: AA Sınıf Geçisi (2 feature)
- F: Mevcut NB32 FE (4 feature)

Toplam deney sayısı: 37

1. Exp 1: Feature Group Ablation

Her feature grubu NB32 baseline FE (Grup F) üzerine eklenerek test edildi. CatBoost modeli ve P9_REVERSE_6040 pool sabit tutuldu.

Deney	Boot-F1	Std	Prec	FP	FN	Delta
E1a_GroupA_FCS	0.7200	0.026	0.926	10	8	-0.0099
E1b_GroupB_EKCombo	0.6929	0.046	0.922	10	14	-0.0370
E1c_GroupC_Missing	0.6998	0.055	0.928	9	17	-0.0302
E1d_GroupDE_AAdelta	0.7576	0.045	0.944	7	14	+0.0276
E1e_All_ABCDEF	0.6899	0.035	0.918	11	10	-0.0400
E1f_Baseline_F	0.7300	0.033	0.933	9	8	+0.0000
E1g_NoFE	0.7104	0.042	0.930	9	14	-0.0195
E1h_All_minus_C	0.7410	0.049	0.944	7	16	+0.0110



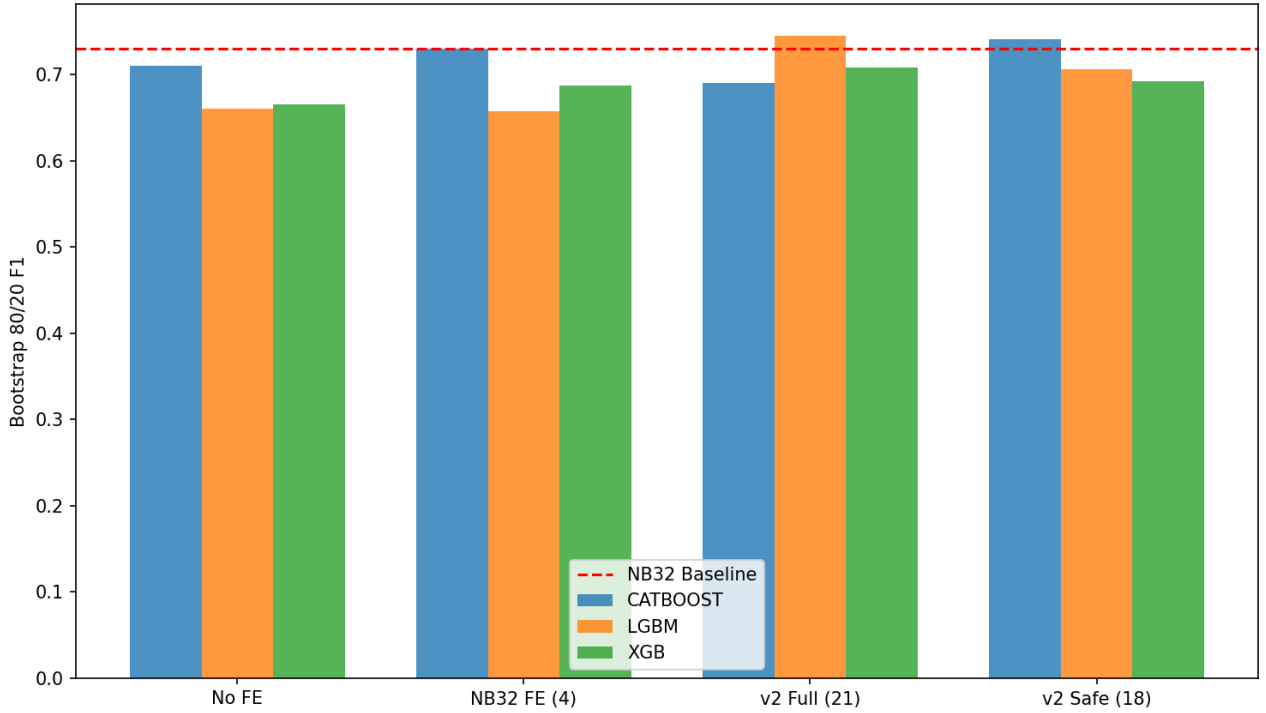
2. Exp 2: FE Versiyon Karsilastirmasi

Uc model tipi (CatBoost, LGBM, XGB) ile dort FE versiyonu karsilastirildi:

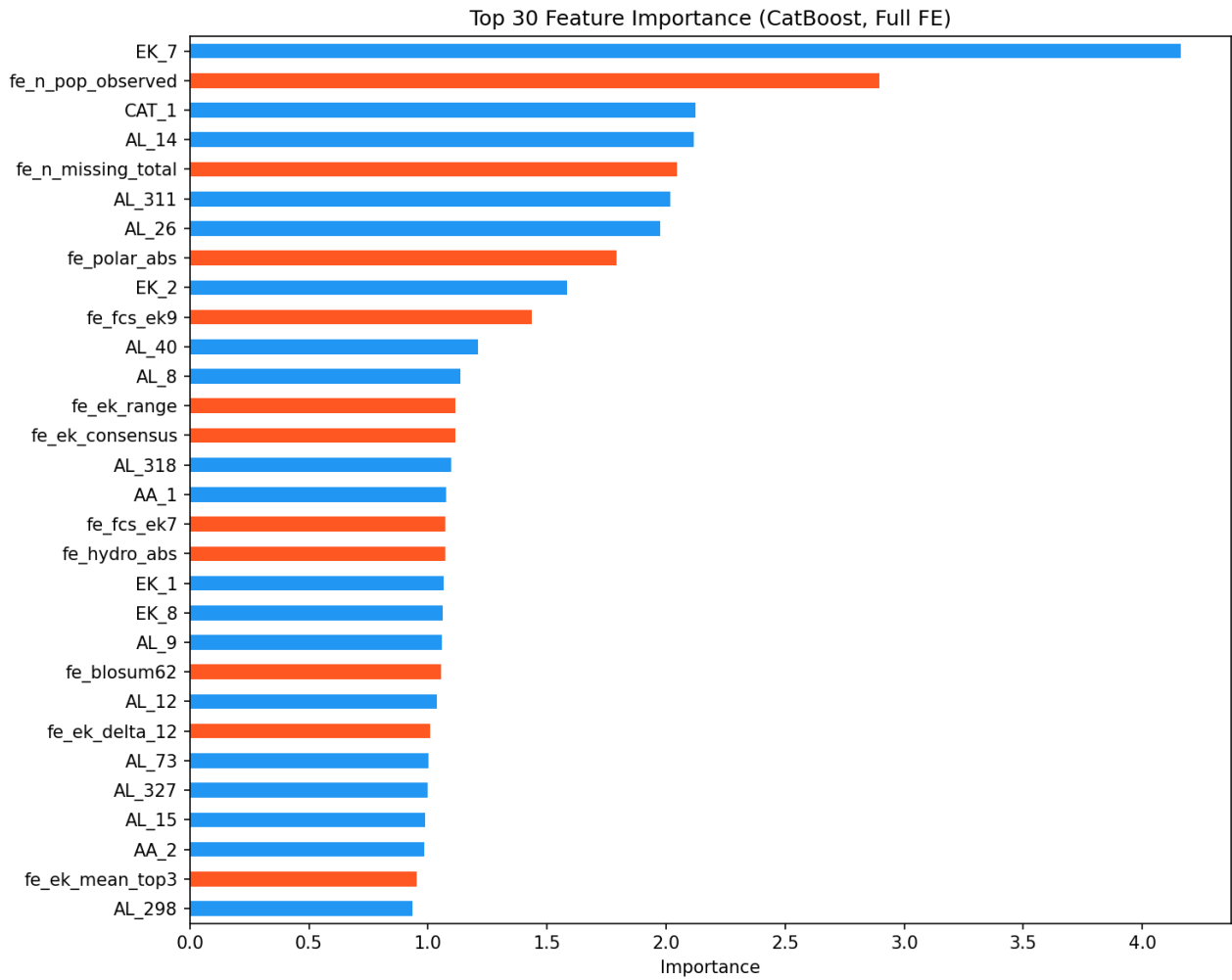
- no_fe: Ham feature'lar
- nb32_fe: NB32 baseline (4 feature)
- v2_full: add_fe_v2 tumu (21 feature)
- v2_safe: Grup C haric (leakage-safe, 18 feature)

Deney	Boot-F1	Std	Prec	FP	FN
catboost/no_fe	0.7104	0.042	0.930	9	14
catboost/nb32_fe	0.7300	0.033	0.933	9	8
catboost/v2_full	0.6899	0.035	0.918	11	10
catboost/v2_safe	0.7410	0.049	0.944	7	16
lgbm/no_fe	0.6601	0.057	0.918	10	21
lgbm/nb32_fe	0.6578	0.058	0.917	10	22
lgbm/v2_full	0.7447	0.047	0.944	7	16
lgbm/v2_safe	0.7065	0.049	0.929	9	15
xgb/no_fe	0.6652	0.054	0.918	10	21
xgb/nb32_fe	0.6876	0.044	0.921	10	16
xgb/v2_full	0.7081	0.046	0.929	9	15
xgb/v2_safe	0.6923	0.056	0.932	8	24

Exp 2: FE Version Comparison



3. Feature Importance (CatBoost)

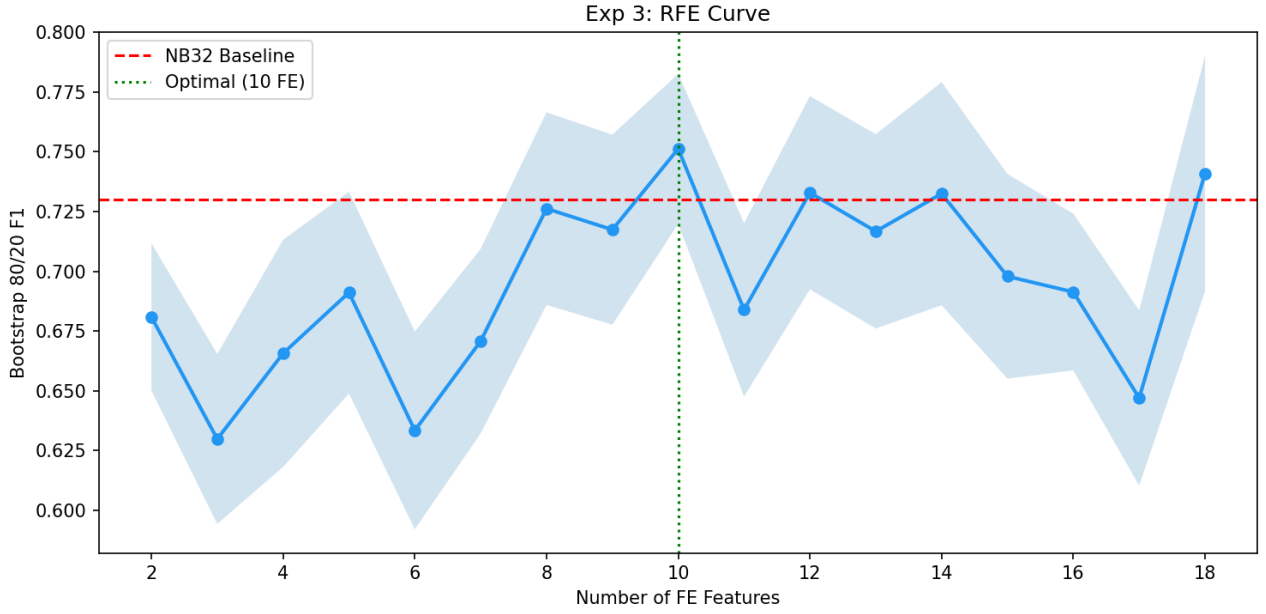


4. Exp 3: RFE - Optimal Feature Subset

Baslangic: 18 FE feature ile RFE.

Optimal: 10 FE feature, Boot-F1=0.7513

Optimal featureler: fe_grantham, fe_blosum62, fe_fcs_ek9, fe_fcs_ek7, fe_ek_mean_top3, fe_ek_max, fe_ek_consensus, fe_ek_delta_12, fe_mw_abs, fe_polar_abs



5. En İyi Model ve Confusion Matrix

En iyi: E1d_GroupDE_AAdelta

Boot-F1: 0.7576 +/- 0.045

Precision: 0.944, FP=7, FN=14

MCC: 0.756, AUPRC: 0.962

NB32 Baseline: 0.7300

Delta: +0.0276

